Human Value Integrated Agent using the Schwartz Value Framework

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1. INTRODUCTION

Artificial intelligence systems are increasingly deployed in domains such as governance, recommendation, content moderation, hiring, and policy support. Although modern AI can optimise for efficiency and predictive accuracy, most systems do not explicitly account for human values such as fairness, autonomy, safety, responsibility, and social welfare. As a result, technically correct outputs may still be socially or ethically undesirable.

To address this gap, we develop a **Human Value Integrated Agent** that incorporates value priorities directly into its decision-making pipeline. We ground the design in the **Schwartz Value Framework**, a well-established psychological theory that organises human values into motivational categories including Benevolence, Power, Security, Self-direction, Universalism, Conformity, Tradition, and Achievement.

The central insight is that different individuals or institutions prioritise values differently, so the same scenario may lead to different but equally valid decisions. Our goal is to build an interpretable, value-conditioned AI system whose choices are transparent and configurable at the level of value preferences.

2. PROBLEM DEFINITION

Given a policy scenario with multiple candidate actions, the task is to build an agent that:

1. Understands the value implications of each action from natural language.
2. Represents internal preferences using Schwartz value dimensions.
3. Selects the action that best aligns with those preferences.
4. Generates a natural-language justification for the chosen action.

This problem is important because real-world decisions involve genuinely competing values:

- **Security vs. Privacy:** surveillance improves safety but erodes civil liberties.
- **Power vs. Fairness:** automation creates wealth but worsens inequality.
- **Profit vs. Welfare:** loosening environmental rules boosts output but harms future generations.
- **Regulation vs. Freedom:** strong rules promote order but limit individual autonomy.

Traditional classifiers predict value labels but do not act as decision-making agents. Purely prompt-based LLMs reason about values opaquely and inconsistently. Our objective is to combine *value understanding* with *structured action selection and transparent justification* in a single modular framework.

3. PROPOSED SOLUTION

Our framework consists of four main modules. The overall system architecture is illustrated in Figure 1.

3.1 Value Prediction

We use the Human Values dataset, where each argumentative sample is annotated with 20 multi-label Schwartz categories. Each example consists of a *Conclusion*, a *Stance*, and a *Premise*, concatenated as a single input string. We evaluate two models:

Baseline: Sentence embeddings from `all-MiniLM-L6-v2` fed into a One-vs-Rest Logistic Regression classifier trained for each of the 20 labels independently.

Improved model: DeBERTa-v3-small fine-tuned end-to-end for multi-label classification with a sigmoid output layer, using BCEWithLogitsLoss, AdamW optimiser ($lr = 2 \times 10^{-5}$), and a linear warmup schedule over 5 epochs.

3.2 Value Aggregation

The 20 fine-grained labels are grouped into 12 interpretable Schwartz dimensions. For each group, the aggregated score is the mean of its constituent label probabilities:

- **Benevolence** = mean(caring, dependability)
- **Power** = mean(dominance, resources)
- **Security** = mean(personal, societal)
- **Conformity** = mean(rules, interpersonal)
- **Universalism** = mean(concern, nature, tolerance, objectivity)
- Self-direction, Tradition, Achievement, Stimulation, Hedonism, Humility, and Face map directly from single labels.

3.3 Agent Profiles

We define four agents, each as a weight vector $\mathbf{w} \in [0, 1]^{12}$ over the Schwartz dimensions:

- **Benevolence Agent:** Benevolence (1.0), Universalism (1.0), Humility (0.8) - welfare-oriented, fairness-driven.
- **Power Agent:** Power (1.0), Achievement (1.0), Face (0.95) - profit- and authority-driven.
- **Security Agent:** Security (1.0), Conformity (1.0), Tradition (0.9) - risk-averse, rules-based.
- **Openness Agent:** Self-direction (1.0), Stimulation (0.9), Universalism (0.9) - autonomy- and exploration-oriented.

3.4 Decision Engine

For a candidate option o with predicted value vector $\mathbf{v}(o)$, alignment with an agent profile $\mathbf{w}$ is calculated as a weighted combination of **Cosine Similarity** and **Spearman Rank Correlation**:

$$\text{Score}(o, \mathbf{w}) = \alpha \cdot \cos(\mathbf{w}, \mathbf{v}(o)) + \beta \cdot \rho(\mathbf{w}, \mathbf{v}(o)) \quad (1)$$

Where α and β are hyperparameters tuned via grid search to maximise decision accuracy. The option $o^* = \arg\max_o \text{Score}(o, \mathbf{w})$ is selected.
Mistral-7B-Instruct (4-bit NF4 quantised via BitsAndBytes) is then prompted with the agent identity, top value priorities, chosen option, and alignment score to generate a persuasive policy justification.

4. EXPERIMENTS

4.1 Dataset

We use the Human Values dataset with the following split:

- Training samples: 5,393
- Validation samples: 1,896
- Label space: 20 binary Schwartz value categories

The dataset is multi-label and class-imbalanced; Macro F1 is therefore an important metric for minority label performance.

4.2 Classification Performance

Table 1 compares the two value-prediction models on the validation split. Micro F1 reflects overall accuracy; Macro F1 gives equal weight to each label regardless of frequency.

Table 1: Comparison of value prediction models.

Model	Micro F1	Macro F1
MiniLM + Logistic Regression	0.4526	**0.4073**
DeBERTa-v3 (fine-tuned)	**0.5380**	0.3790
RoBERTa-base (fine-tuned)	**0.5698**	**0.4494**

4.3 Scenario-Based Agent Evaluation

We evaluate the full pipeline on three policy scenarios. To ensure interpretability, each scenario is designed with three distinct ethical archetypes:

- **Option A** (Centralized/Dominant),
- **Option B** (Moderate/Balanced),
- and **Option C** (Individual/Liberal).

Table 2 reports the decisions made by each agent using the tuned DeBERTa-v3 pipeline.

Table 2: Agent decisions across policy scenarios. (A: Pro-Authority, B: Balanced, C: Pro-Liberty).

Scenario	Ben.	Pow.	Sec.	Open.
Privacy vs State Control	B	A	B	C
Automation Transition	B	A	C	B
Environmental Regulation	C	A	B	C

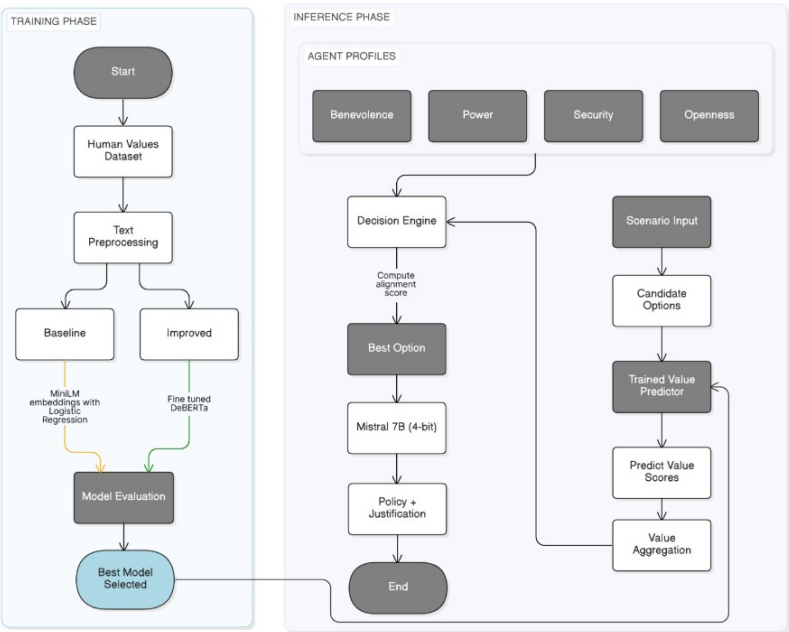

Figure 1: System architecture: Training Phase (left) and Inference Phase (right). The training phase produces a fine-tuned DeBERTa value predictor; the inference phase uses agent profiles and the trained predictor to score candidate options and generate policy justifications via Mistral-7B.

5. RESULTS & OBSERVATIONS

5.1 Value Prediction Results

DeBERTa achieves the best Micro F1 of 0.5380, outperforming the MiniLM + Logistic Regression baseline (0.4526) by 8.5 percentage points, confirming that contextual transformer representations capture nuanced value semantics more effectively than static sentence embeddings. The baseline yields a slightly higher Macro F1 (0.4073 vs. 0.3790), indicating more balanced recall on minority labels - an expected trade-off from DeBERTa's tendency to favour high-frequency labels.

5.2 Model Comparison and Hyperparameter Tuning

We performed a grid search to find the optimal α and β for both transformer architectures. Results are summarized in Table 3.

Table 3: Comparative analysis of decision logic and stability.

Metric / Parameter	DeBERTa-v3	RoBERTa-base
Optimal Alpha (α)	0.25	0.75
Optimal Beta (β)	0.00	0.25
Training Pass Rate	**91.4%**	77.1%
Held-out Pass Rate	**60.0%**	46.7%
Power Agent Stability	Stable (Opt A)	Moderate (Opt B)

Interpretation: DeBERTa-v3 demonstrates superior stability, achieving peak accuracy with $\beta = 0$. This indicates that its raw value intensities are reliable enough to drive decisions without relying on rank correlation. In contrast, RoBERTa-base requires $\beta = 0.25$, suggesting its individual value scores are noisier and require the relative ordering of values to reach a correct decision.

5.3 The Generalization Gap

While both models perform well on training scenarios, we observe a "Generalization Gap" in the held-out tests (60% for DeBERTa vs 46.7% for RoBERTa). This suggests that while the agent logic is sound, the underlying value predictors are sensitive to the specific linguistic domain of the scenario.

5.4 Value-Conditioned Decision Behaviour

Identical scenarios produce systematically different, value-coherent decisions depending on the agent profile:

- The **Power Agent** consistently selects profit-maximising options (Option A in all three scenarios).
- The **Security Agent** favours cautious, stability-oriented policies (Option B or C).
- The **Openness Agent** selects privacy- and freedom-preserving options (Option C in two of three scenarios).
- The **Benevolence Agent** selects socially supportive or balanced solutions (Options B and C).

This demonstrates successful value conditioning: the same inference pipeline produces diverse, value-coherent outputs purely through profile weighting, without any retraining.

5.5 Generated Justifications

Mistral-7B-Instruct generates coherent policy statements aligned with each agent's priorities. For the Automation Transition scenario:

- **Power Agent** emphasised market competitiveness and rapid adoption for economic leadership.
- **Benevolence Agent** centred on worker welfare, retraining, and safety nets for an equitable transition.
- **Security Agent** argued for gradual, phased implementation to preserve employment stability.
- **Openness Agent** framed the argument around human-centred, inclusive growth with worker agency.

5.6 Interpretability

Unlike end-to-end black-box prompting, our pipeline explicitly separates three auditable stages: (i) value prediction, (ii) alignment scoring, and (iii) language generation. A practitioner can inspect the predicted value vector, agent weights, and per-option alignment scores to understand exactly why a particular option was chosen. Changing a single weight in the agent profile changes the decision without retraining, providing direct post-hoc controllability.

6. FUTURE SCOPE

Learned Agent Profiles. Current profiles are manually specified weight vectors. Future work can learn personalised profiles from user behaviour or preference feedback, allowing adaptation to individual or organisational priorities.

Improved Value Classification. Rare value labels remain harder to predict. Class-weighted training, data augmentation, or stronger models could improve minority-label performance.

Domain Extension. The framework can be extended to real-world domains such as healthcare, education, legal reasoning, and public policy, where value-aware decision making is especially important.

CODE AVAILABILITY

The implementation of the DeBERTa-based value predictor, agent alignment logic, and the Mistral-7B justification pipeline is open-source and available at:
https://github.com/Khushbu0405/human-value-integrated-agent

7. CONCLUSION

We presented a Human Value Integrated Agent that combines value prediction, profile-based decision making, and LLM-based justification generation in a unified framework. Experimental results show that DeBERTa improves semantic value prediction, while agent profiles successfully control downstream decisions across multiple scenarios. The generated justifications further demonstrate the ability of the system to produce coherent and value-aligned policy explanations. Overall, the framework provides a transparent and configurable approach for aligning AI systems with diverse human values.

References

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